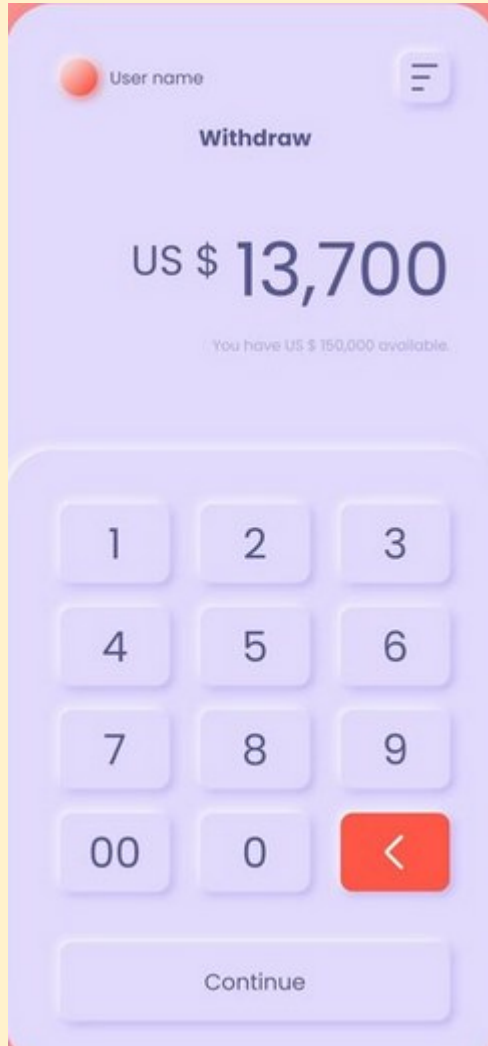


Intro to Android Studio and Mobile App



App views



Withdraw

User name 

US \$ 13,700

You have US \$ 150,000 available.

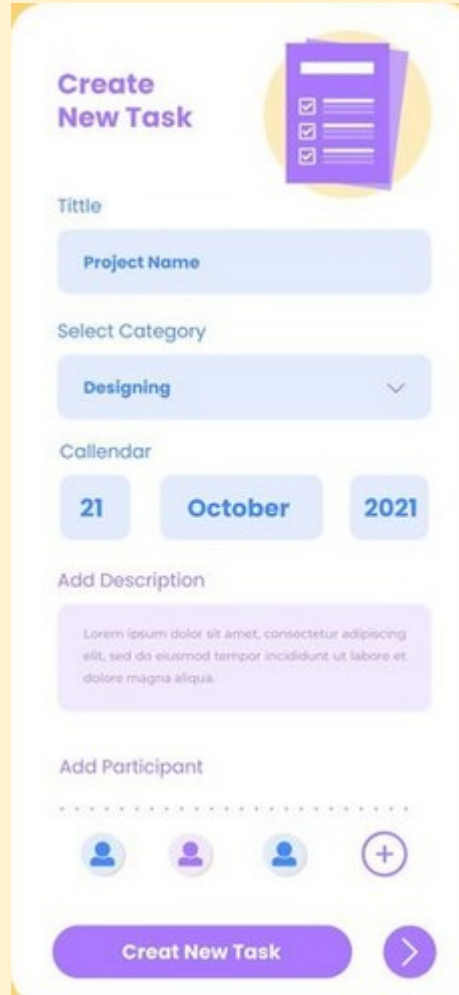
1 2 3

4 5 6

7 8 9

00 0 <

Continue



Create New Task 

Title

Project Name

Select Category

Designing

Calendar

21 October 2021

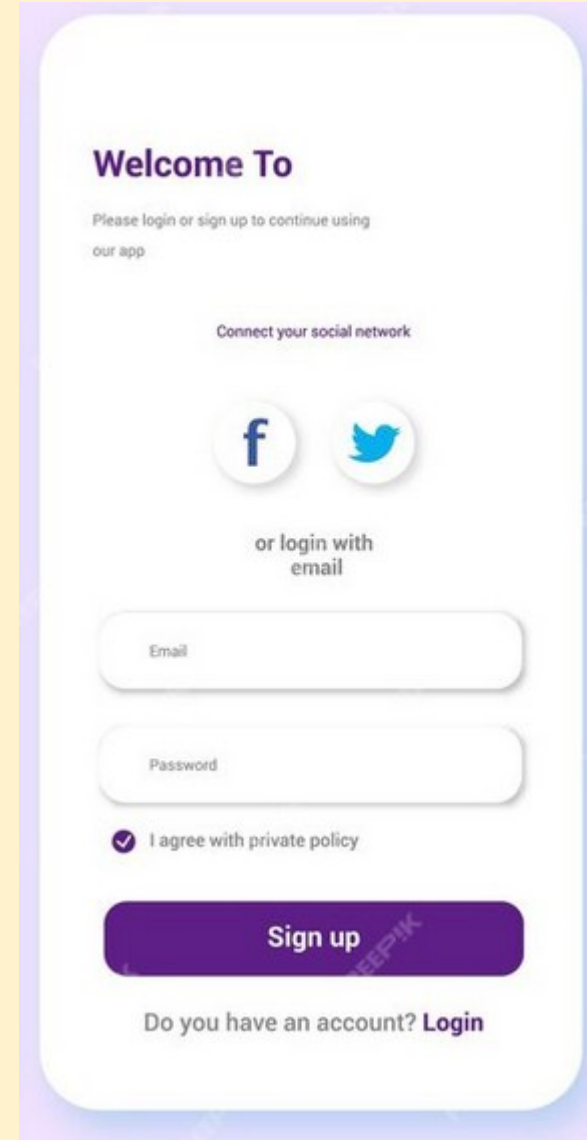
Add Description

Lorem ipsum dolor sit amet, consectetur adipiscing elit, sed do eiusmod tempor incididunt ut labore et dolore magna aliqua.

Add Participant



Creast New Task



Welcome To

Please login or sign up to continue using our app

Connect your social network

or login with email

Email

Password

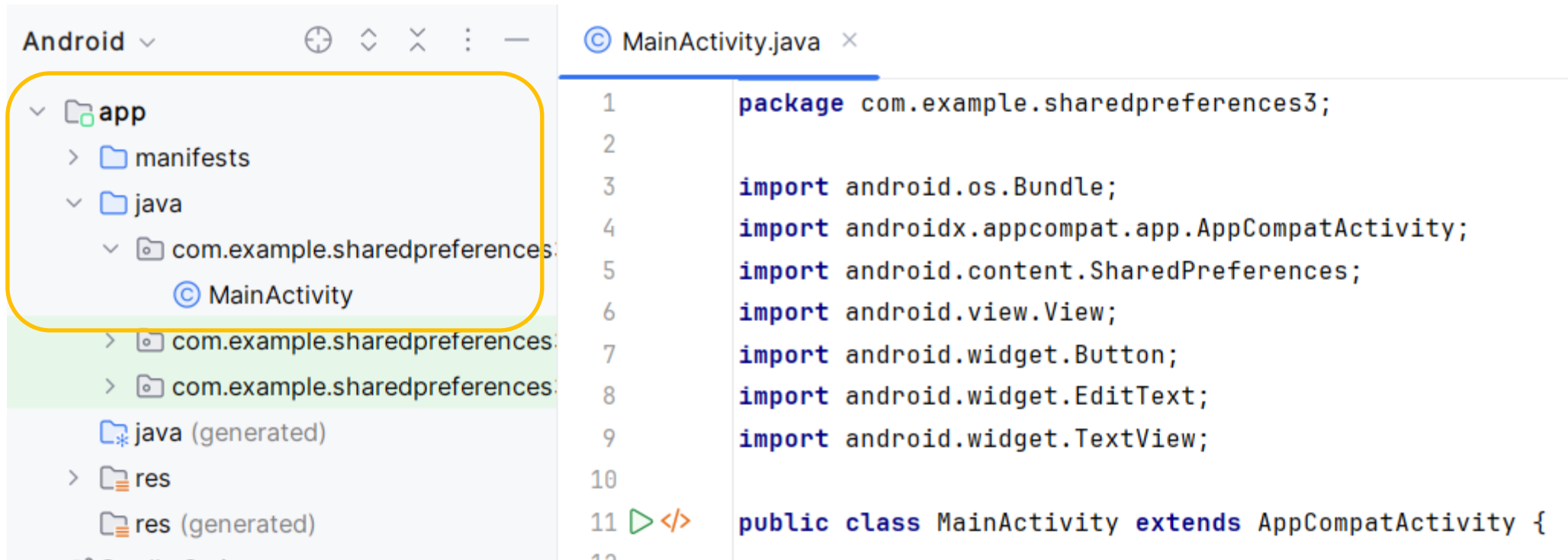
☒ I agree with private policy

Sign up

Do you have an account? Login

What's in an app? (application components)

Activities: An activity is the key execution component . You need at least one activity in an app that deals with the UI.

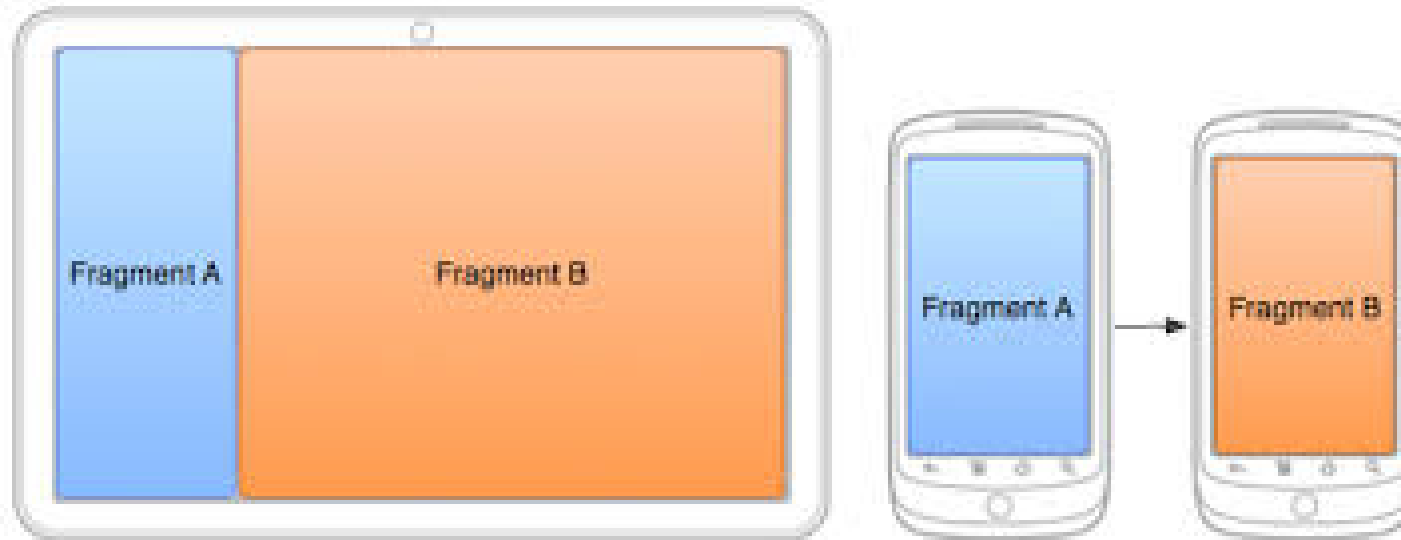


The screenshot displays the Android Studio interface. On the left, the 'Android' tab shows the project structure. The 'app' folder is expanded, revealing 'manifests', 'java', and 'res' subfolders. The 'java' folder is further expanded, showing the package 'com.example.sharedpreferences3' and the 'MainActivity' class, which is highlighted with a yellow circle. On the right, the 'MainActivity.java' file is open, showing the following code:

```
1 package com.example.sharedpreferences3;
2
3 import android.os.Bundle;
4 import androidx.appcompat.app.AppCompatActivity;
5 import android.content.SharedPreferences;
6 import android.view.View;
7 import android.widget.Button;
8 import android.widget.EditText;
9 import android.widget.TextView;
10
11 public class MainActivity extends AppCompatActivity {
12
```

What's in an app? (application components)

Fragments: Fragments bring more flexibility in handling app functionalities and user interfaces.



What's in an app? (application components)

Services: Services are long running programs that don't need to interact with the UI . Examples of services could be listening to music, updating the location. An activity will control a service; that is, start it, pause it, bind and get data from it.



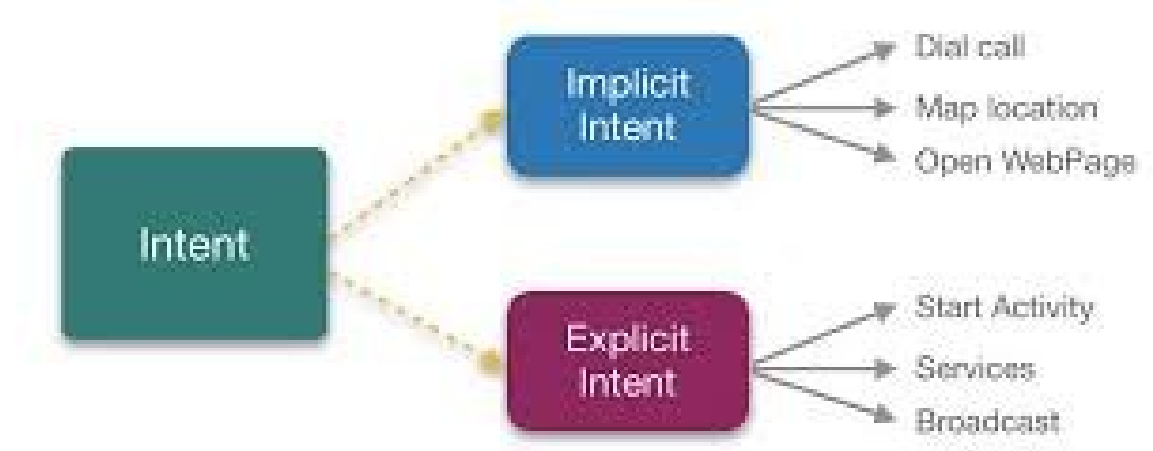
What's in an app? (application components)

Broadcast Receivers: They handle communication between Android OS and applications.

Content Providers: They handle data and database management issues.

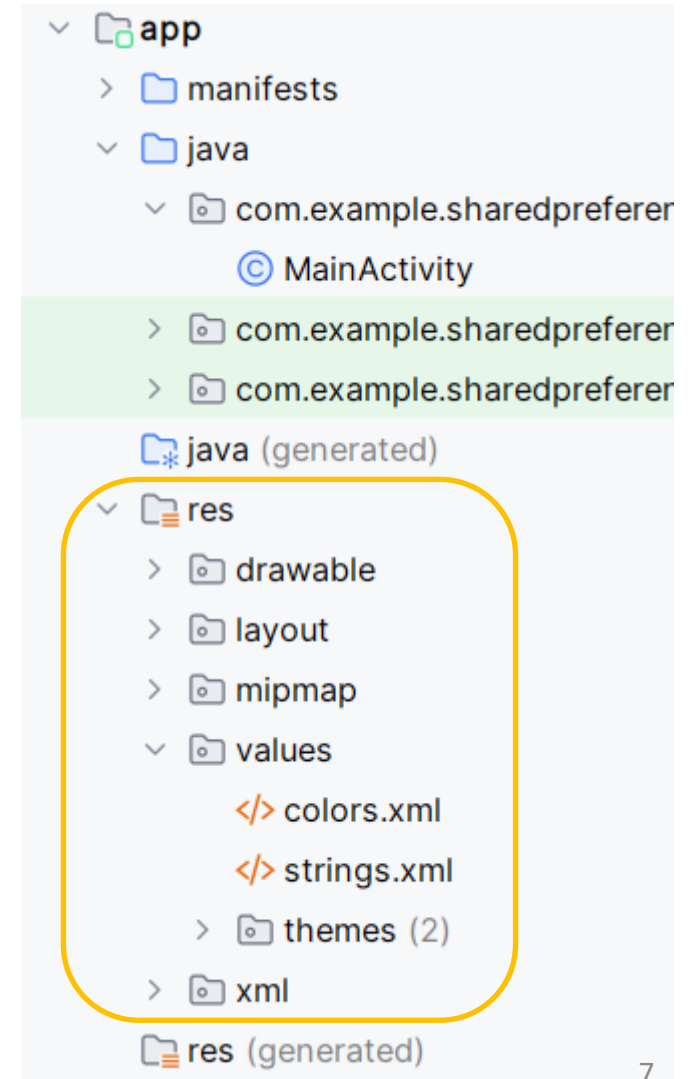
Manifest+Res

Intent: Explicit and implicit.



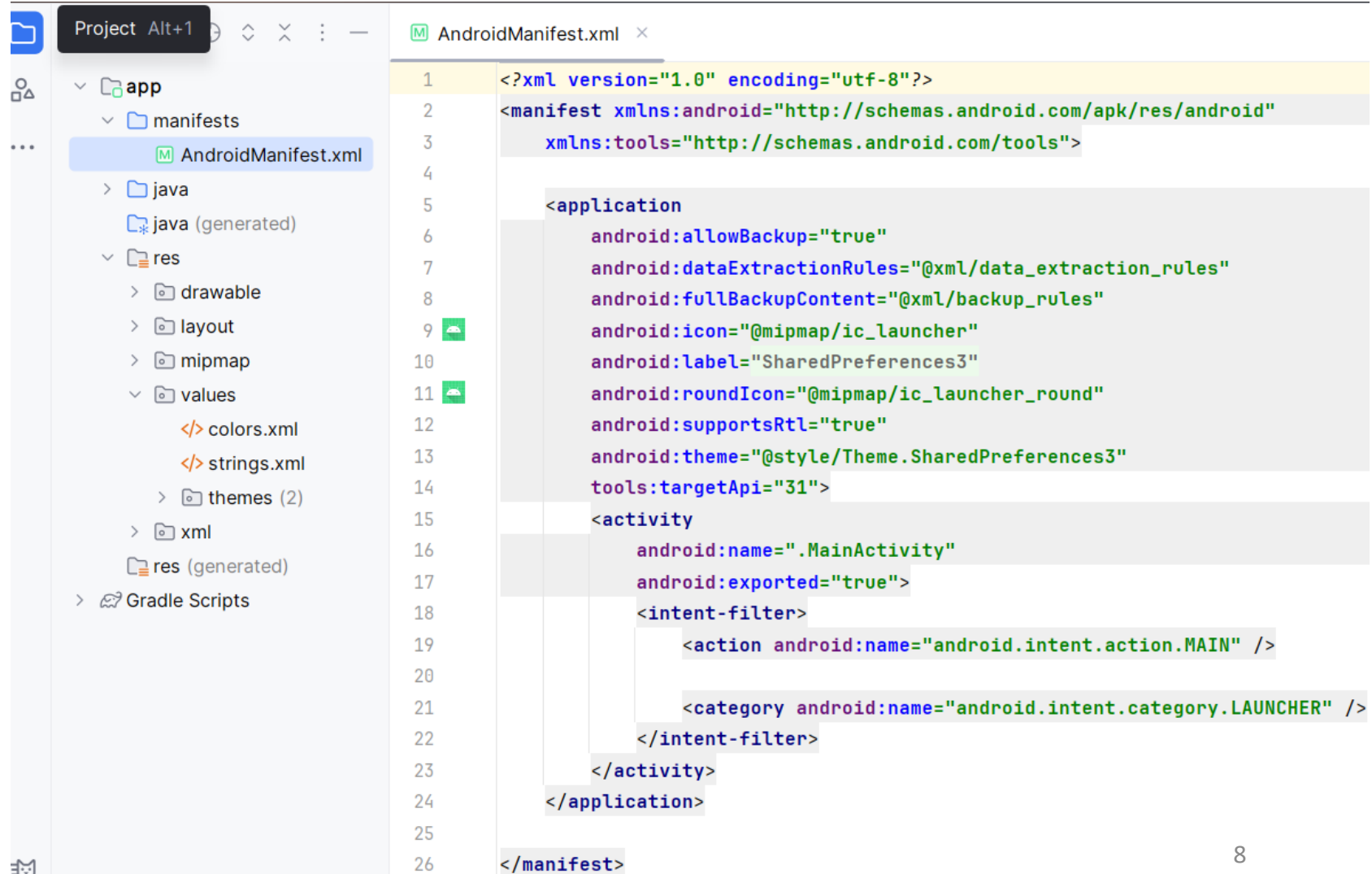
What's in an app? (application components)

Resources: External elements, such as strings, constants and drawable pictures.



What's in an app? (application components)

Manifest:



```
1 <?xml version="1.0" encoding="utf-8"?>
2 <manifest xmlns:android="http://schemas.android.com/apk/res/android"
3         xmlns:tools="http://schemas.android.com/tools">
4
5     <application
6         android:allowBackup="true"
7         android:dataExtractionRules="@xml/data_extraction_rules"
8         android:fullBackupContent="@xml/backup_rules"
9         android:icon="@mipmap/ic_launcher"
10        android:label="SharedPreferences3"
11        android:roundIcon="@mipmap/ic_launcher_round"
12        android:supportsRtl="true"
13        android:theme="@style/Theme.SharedPreferences3"
14        tools:targetApi="31">
15
16        <activity
17            android:name=".MainActivity"
18            android:exported="true">
19            <intent-filter>
20                <action android:name="android.intent.action.MAIN" />
21
22                <category android:name="android.intent.category.LAUNCHER" />
23            </intent-filter>
24        </activity>
25    </application>
26</manifest>
```

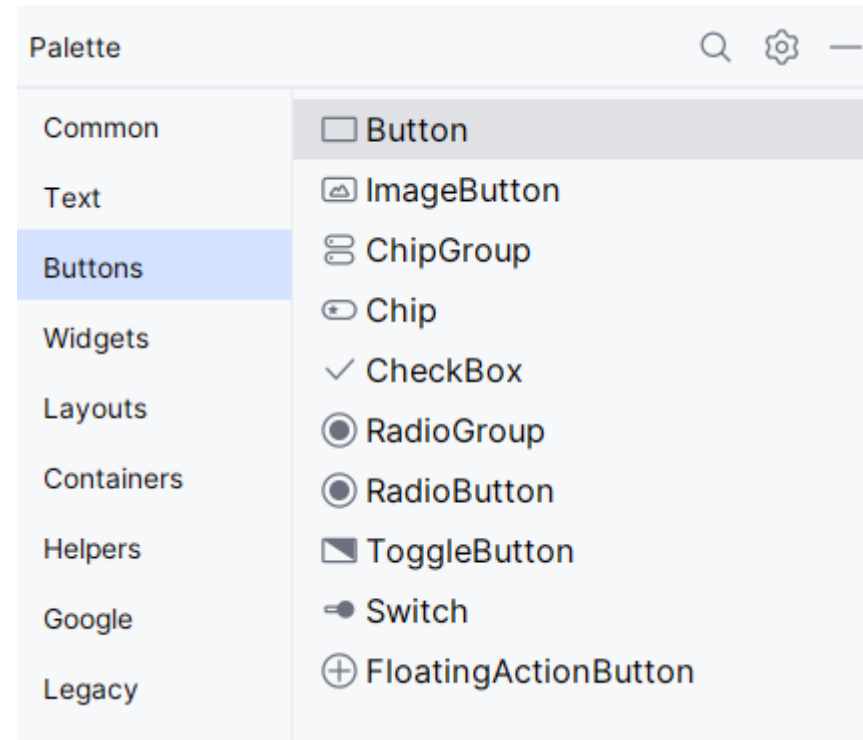
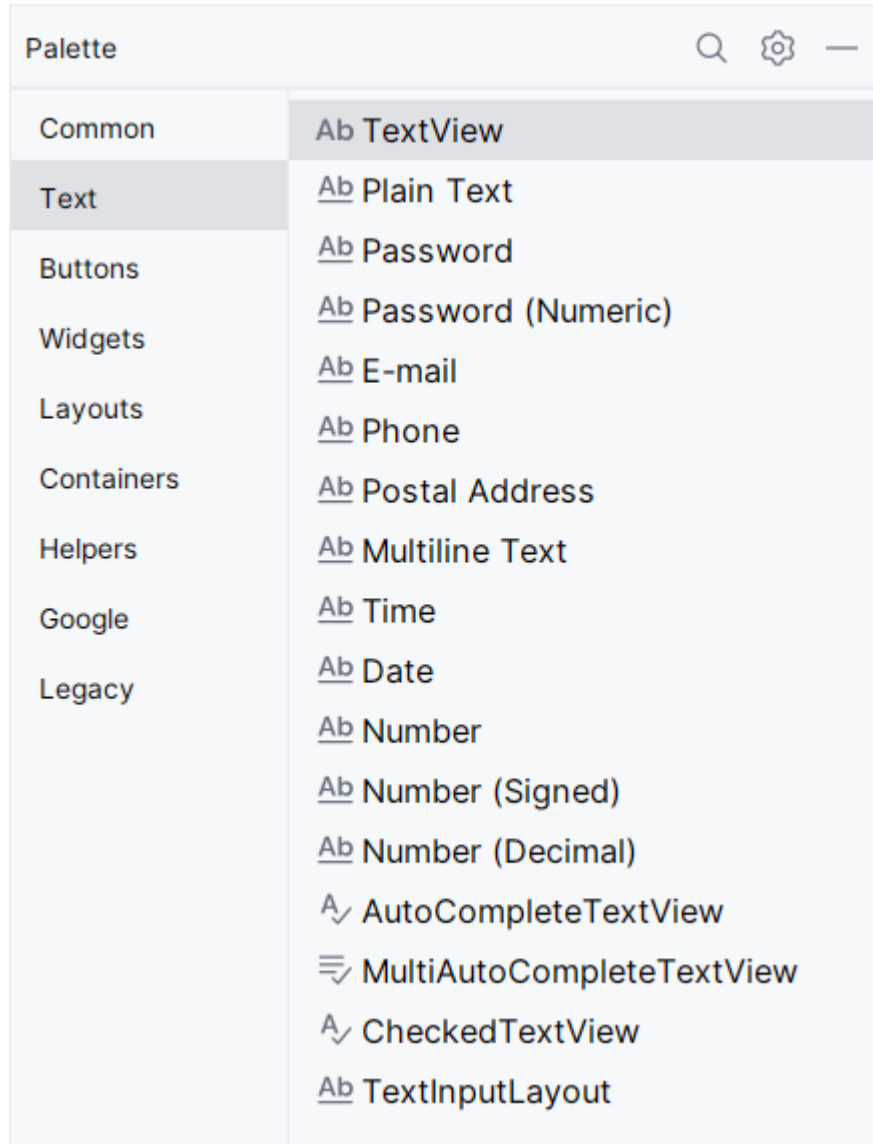

What's in an app? (application components)

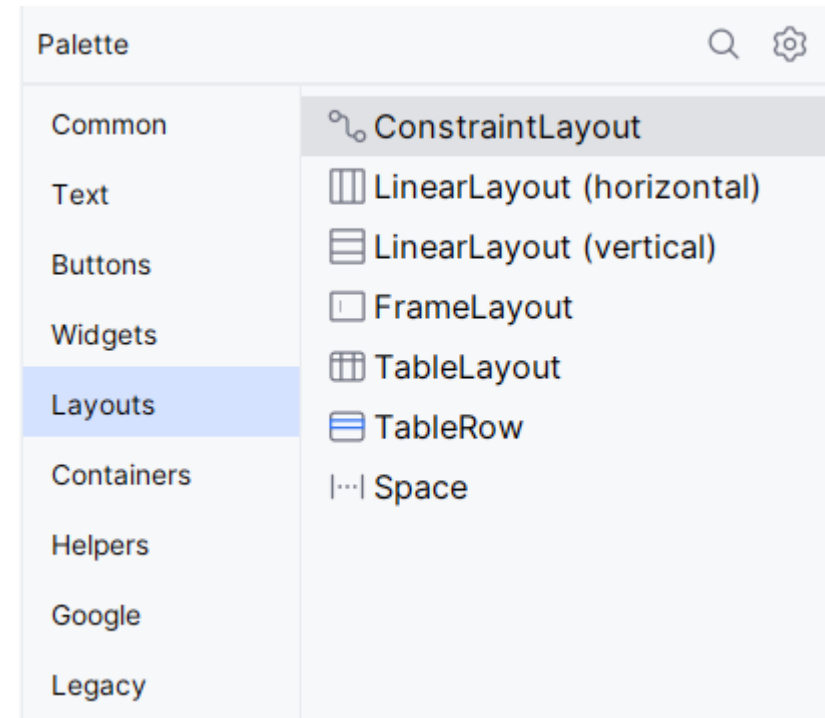
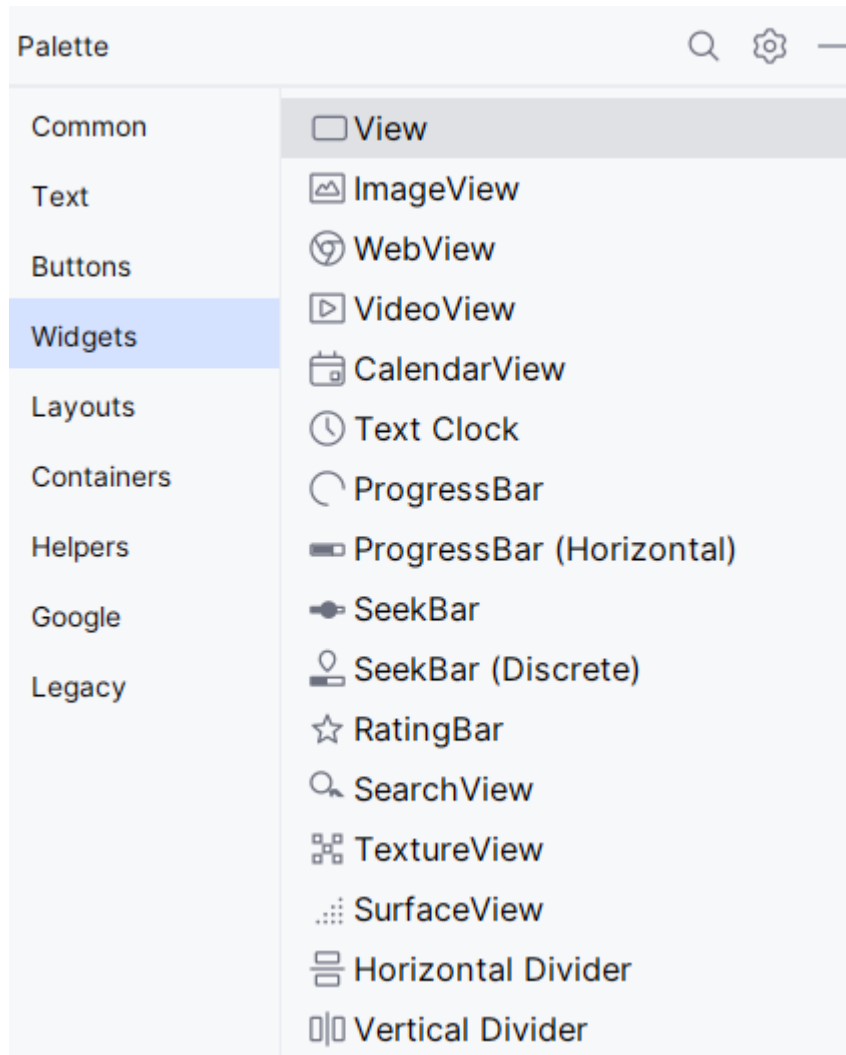
View: UI elements that are drawn on-screen.

Types of views:

- TextView
- EditText
- Button
- Image Button
- Date Picker
- RadioButton
- CheckBox buttons
- Image View







Palette

Common

Text

Buttons

Widgets

Layouts

Containers

Helpers

Google

Legacy

Spinner

RecyclerView

ScrollView

HorizontalScrollView

NestedScrollView

ViewPager2

CardView

AppBarLayout

BottomAppBar

NavigationView

BottomNavigationView

Toolbar

MaterialToolbar

TabLayout

TabItem

ViewStub

ViewAnimator

ViewSwitcher

<include>

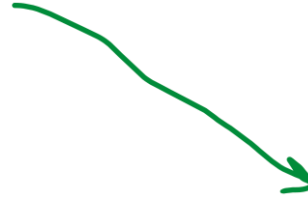
FragmentContainerView

NavHostFragment

<view>

<requestFocus>

Tree



jay@gmail.com

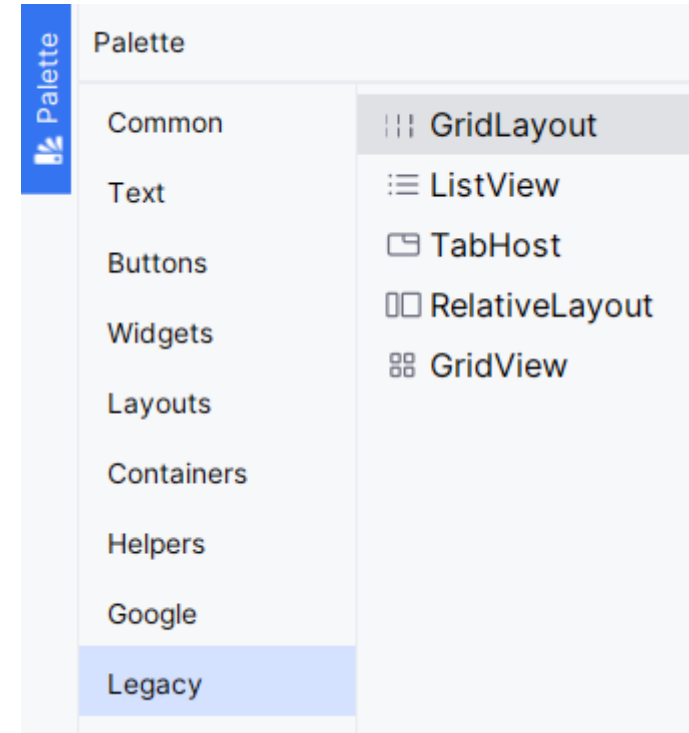
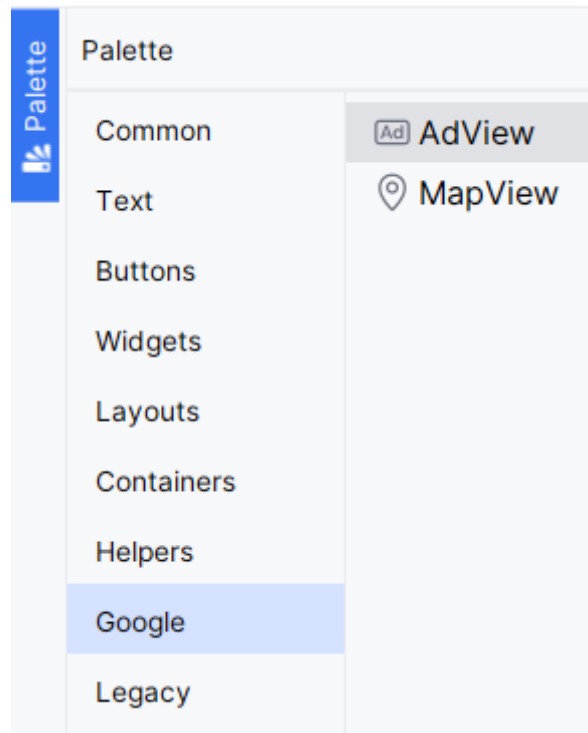
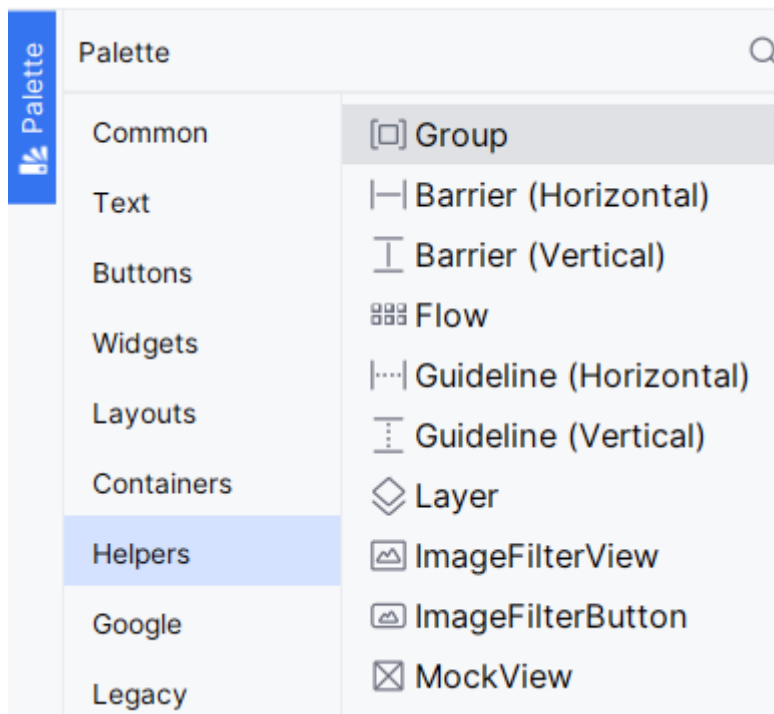
Home

Home

Work

Other

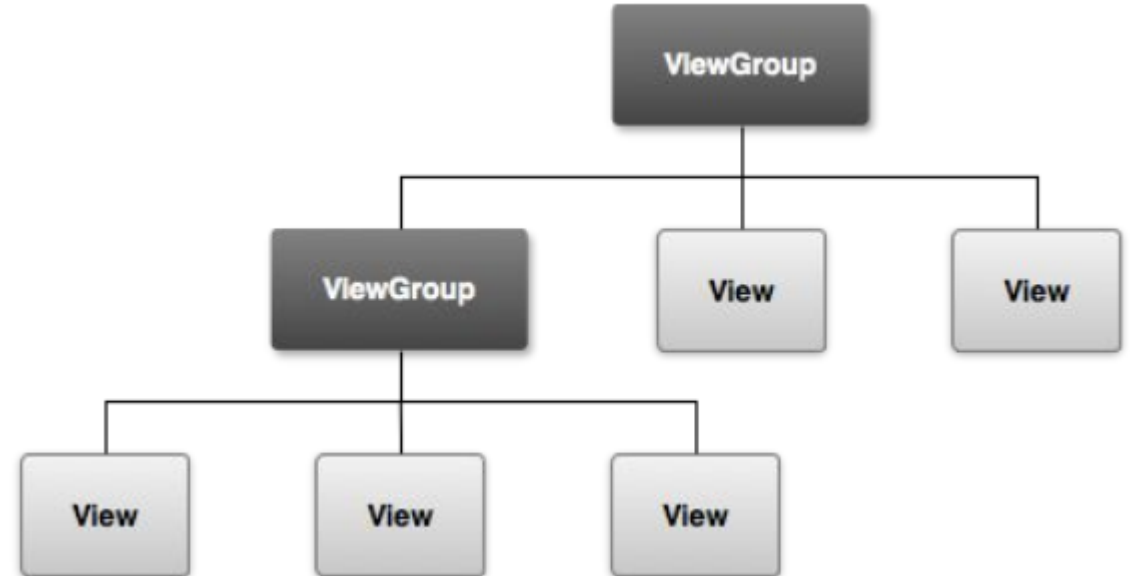
Custom



ViewGroup

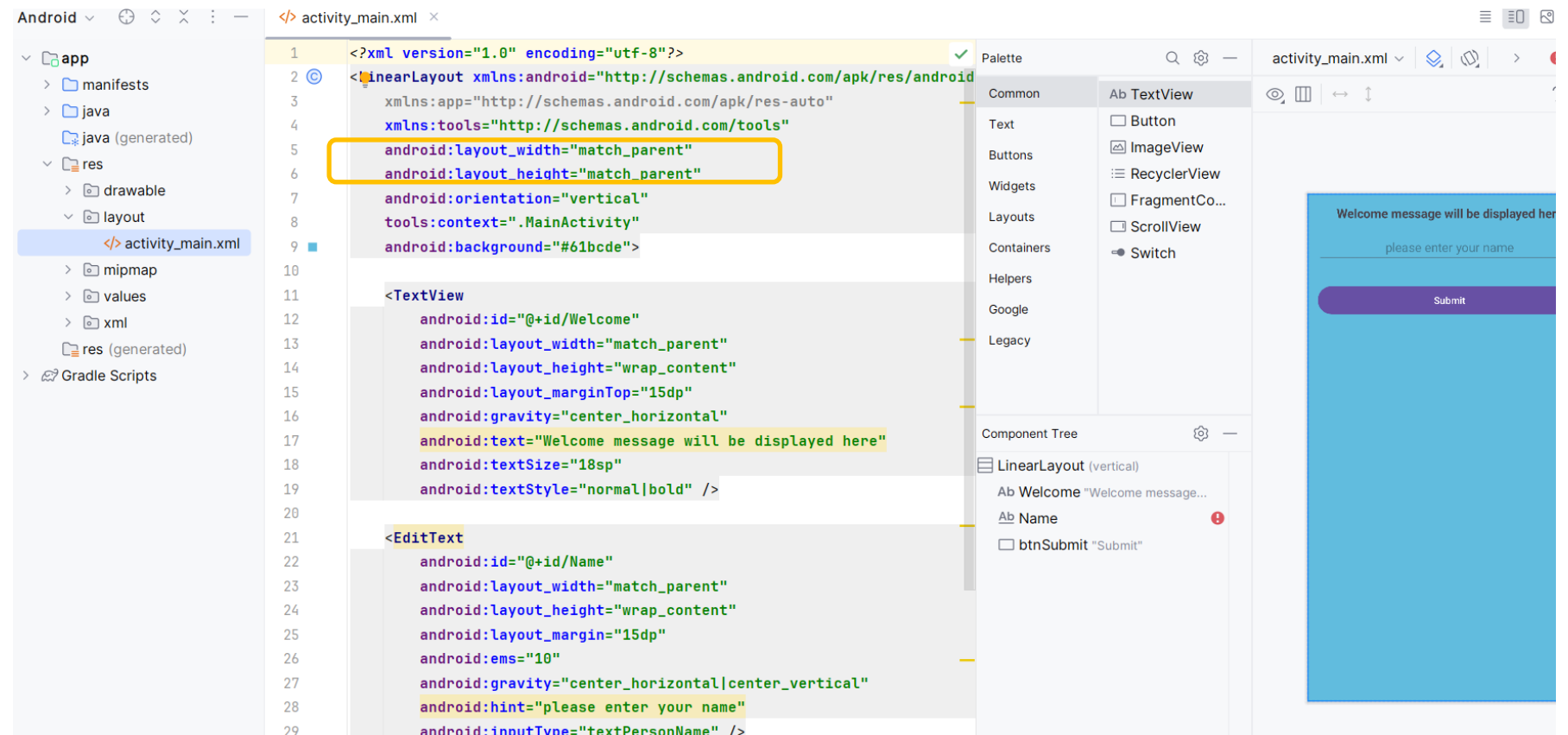
A **View Group** is a subclass of the ViewClass and can be considered as a superclass of Layouts. It provides an **invisible container to hold the views or layouts**. ViewGroup instances and views work together as a container for Layouts.

- LinearLayout
- RelativeLayout
- FrameLayout
- GridView
- ListView



What's in an app? (application components)

Layout: View hierarchies that control screen format and appearance of the views.



Common layout



RelativeLayout



GridLayout

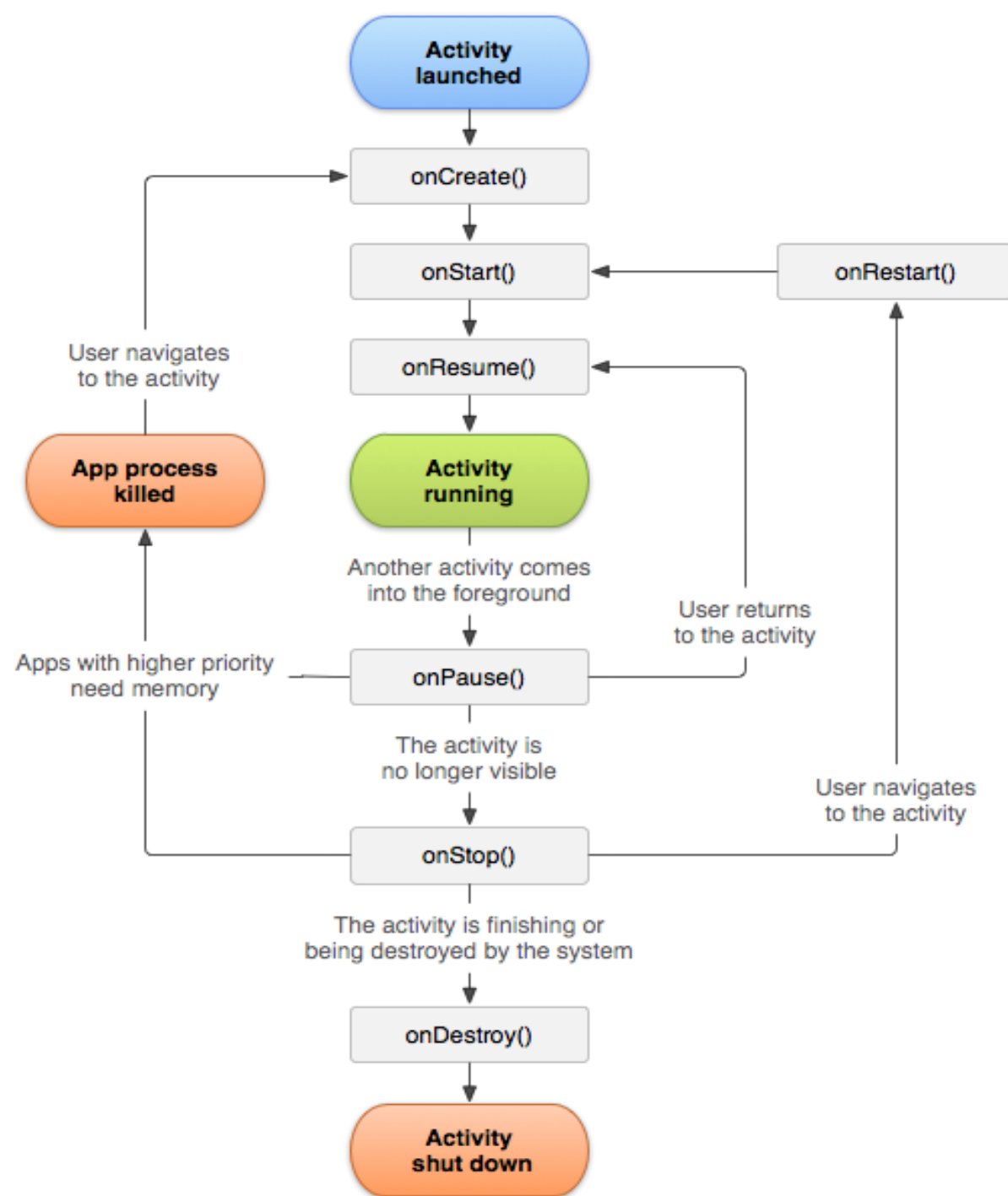


ScrollView

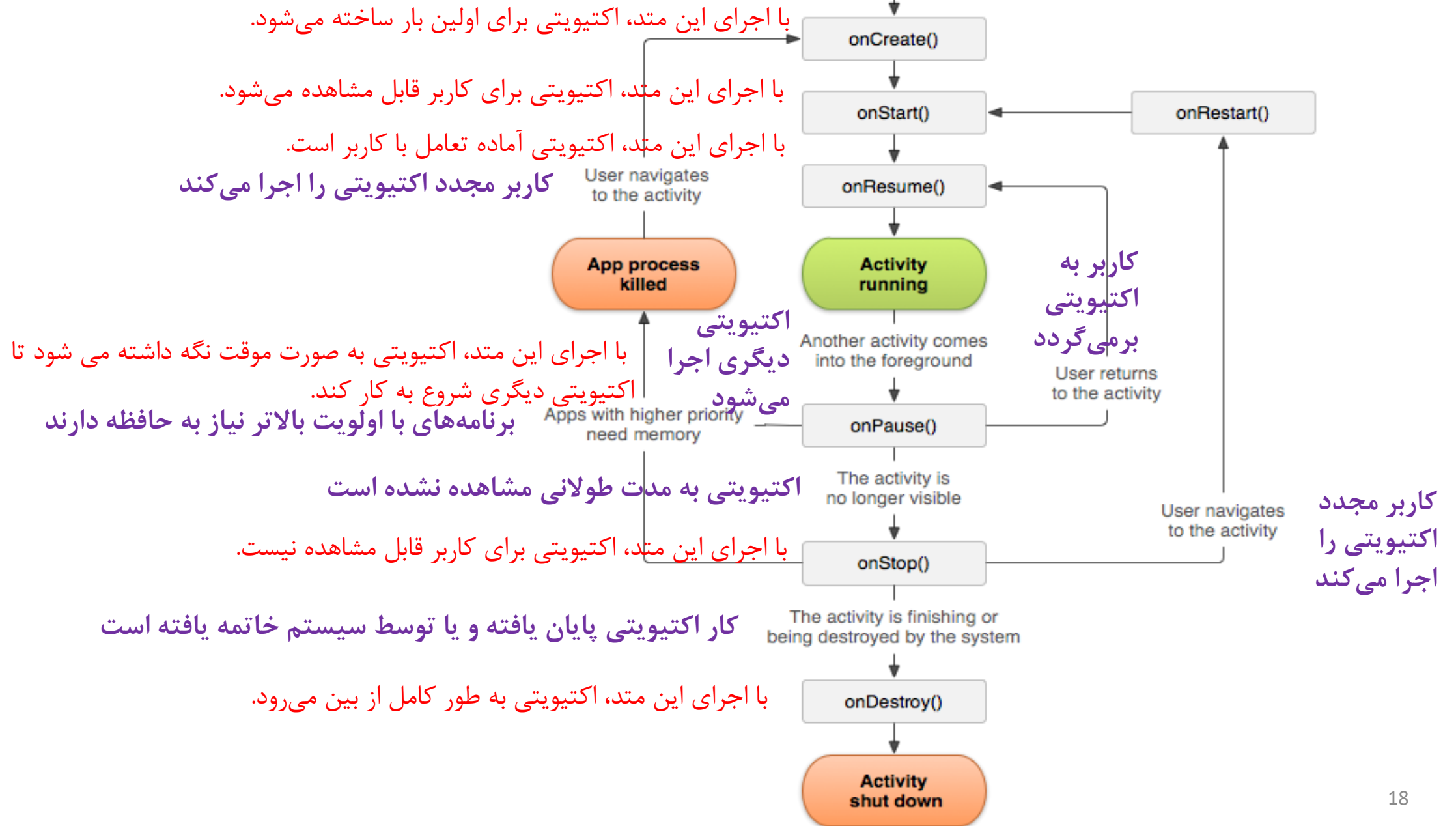


Frame

Activity Lifecycle



Activity Lifecycle



Create a project

Templates

Phone and Tablet

Wear OS

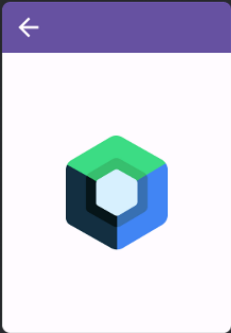
Television

Automotive

New Project



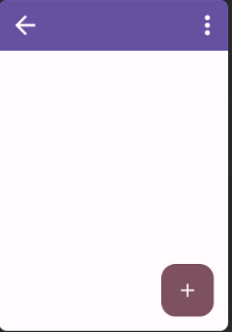
No Activity



Empty Activity



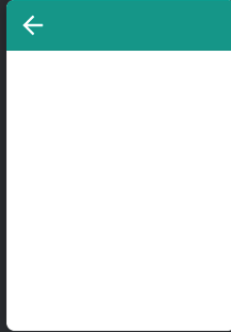
Gemini API Starter



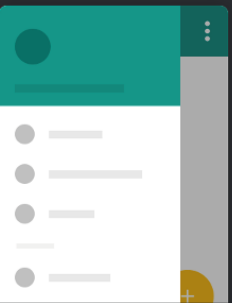
Basic Views Activity

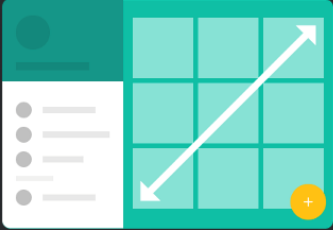


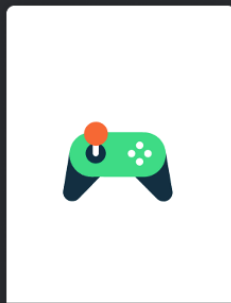
Bottom Navigation Views Activity



Empty Views Activity







Cancel

Previous

Next

Finish

20

Basic Activity

Creates a new basic activity with the Navigation component

Name

My Application

Package name

com.example.myapplication

Save location

/Users/User/AndroidStudioProjects/MyApplication

Language

Kotlin

In this course, we choose Java.

Minimum SDK

API 21: Android 5.0 (Lollipop)

 Your app will run on approximately **98.8%** of devices.
[Help me choose](#)

☐ Use legacy android.support libraries 
Using legacy android.support libraries will prevent you from using
the latest Play Services and Jetpack libraries

Cancel

Previous

Next

Finish

Java Program

```
class HelloWorldApp {  
    public static void main(String[] args) {  
        System.out.println("Hello World!");  
    }  
}
```

HelloWorldApp.java

Compiler

The output of the compiler is **.class** file
bytecode

Interpreter

Interpreter

Interpreter

machine code



Win32



Solaris



MacOS

The Interpreter's are sometimes referred to as the Java Virtual Machines (JVM)

